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PHOTOELECTRIC CONVERSION DEVICE AND APPARATUS

Abstract

A photoelectric conversion device is provided. The device includes pixels each including first and second converters and a row selector. A holding unit and an AD converter are arranged for each column. The holding unit includes a holding circuit to receive signals from pixels via output lines and a selector. The row selector causes at least two pixels that are connected to different output lines and arranged on different rows to output first signals based on a signal of the first converter, and then output second signals based on signals of the first and second converter. The selector selects the first and second signals that are held by the holding circuit and outputs them to the AD converter, and a number of first signals output to the AD converter is smaller than a number of first signals supplied to the holding circuit.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present disclosure relates to a photoelectric conversion device and an apparatus.

Description of the Related Art

[0002] In Japanese Patent Laid-Open No. 2021-153255, high-speed shooting is requested of a photoelectric conversion device that obtains both a focus detection signal and an image generation signal. To achieve speeding up, it is considered to increase the number of rows on which signals are analog-to-digital (A/D)-converted at the same time in one horizontal period, or shorten one horizontal period. Japanese Patent Laid-Open No. 2021-153255 discloses a technique in which some of focus detection data A/D-converted at the same time during one horizontal period are not output so as to shorten one horizontal period.

SUMMARY OF THE INVENTION

[0003] A photoelectric conversion device needs to achieve not only high-speed shooting, but also high-precision focus detection and reduction of power consumption.

[0004] Some embodiments of the present disclosure provide a technique advantageous for improving the performance of a photoelectric conversion device.

[0005] According to some embodiments, a photoelectric conversion device comprising: a plurality of pixels arranged to constitute a plurality of rows and a plurality of columns; a plurality of output lines; and a row selector configured to select, from the plurality of pixels, a pixel that outputs a signal, wherein a signal holding unit, and a converter configured to convert an analog signal output from the signal holding unit into a digital signal are arranged for each column of the plurality of columns, the signal holding unit includes a holding circuit configured to receive signals from pixels connected to at least two output lines out of the plurality of pixels via, out of the plurality of output lines, the at least two output lines assigned to the column on which the signal holding unit is arranged, and a selection circuit, each of the plurality of pixels includes a first photoelectric converter and a second photoelectric converter, the row selector causes at least two pixels that are connected to different output lines out of the at least two output lines and arranged on different rows out of the plurality of rows, to output first analog signals based on a signal of the first photoelectric converter, and then output second analog signals based on a signal of the first photoelectric converter and a signal of the second photoelectric converter, the selection circuit selects the first analog signals and the second analog signals that are held by the holding circuit, and outputs the first analog signals and the second analog signals to the converter, and a number of first analog signals output by the selection circuit to the converter is smaller than a number of first analog signals supplied to the holding circuit, is provided.

[0006] Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a block diagram showing an example of the arrangement of a photoelectric conversion device according to an embodiment;

[0008] FIG. 2 is a schematic view showing an example of the arrangements of the pixel portion and peripheral circuit of the photoelectric conversion device in FIG. 1;

[0009] FIG. 3 is a diagram showing an example of the arrangement of the signal holding unit of the photoelectric conversion device in FIG. 1;

[0010] FIG. 4 is a circuit diagram showing an example of the arrangement of the pixel of the photoelectric conversion device in FIG. 1;

[0011] FIG. 5 is a view showing an example of the arrangement of the pixel of the photoelectric conversion device in FIG. 1;

[0012] FIG. 6 is a block diagram showing an example of the arrangement of the controller of the photoelectric conversion device in FIG. 1;

[0013] FIGS. 7A to 7D are timing charts showing an example of the operation of the photoelectric conversion device in FIG. 1;

[0014] FIGS. 8A to 8D are timing charts showing an example of the operation of the photoelectric conversion device in FIG. 1;

[0015] FIGS. 9A to 9D are timing charts showing an example of the operation of the photoelectric conversion device in FIG. 1; and

[0016] FIG. 10 is a view showing an example of the arrangement of an apparatus incorporating the photoelectric conversion device in FIG. 1.

DESCRIPTION OF THE EMBODIMENTS

[0017] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

[0018] A photoelectric conversion device according to an embodiment of the present disclosure will be explained with reference to FIGS. 1 to 9. FIG. 1 is a block diagram showing an example of the arrangement of a photoelectric conversion device **100** according to the present disclosure. The photoelectric conversion device **100** is controlled by, for example, a CPU **101** arranged outside the photoelectric conversion device **100**. The photoelectric conversion device **100** can include a controller **102**, a row selector **103**, a pixel portion **104**, a column analog-to-digital (A/D) converter **105**, a data processor **106**, and a signal output unit **107**. The controller **102** receives a sync signal and a control signal that are supplied from the CPU **101**, and controls components within the photoelectric conversion device **100**. At the pixel portion **104**, a plurality of pixels **201** are so arranged as to constitute a plurality of rows and a plurality of columns. In the arrangement shown in FIG. 1, the pixels **201** are arranged in a $n \times m$ matrix. In accordance with a control signal supplied from the controller **102**, the row selector **103** selects, from the pixels **201**, a pixel that outputs a signal. In accordance with a control signal supplied from the controller **102**, the column A/D converter **105** converts an analog signal output from the pixel portion **104** into a digital signal. In accordance with a control signal supplied from the controller **102**, the data processor **106** performs signal processing such as correction processing on the signal output from the column A/D converter **105**. The signal output unit **107** can be an interface for outputting a signal to outside the photoelectric conversion device **100**.

[0019] FIG. 2 is a schematic view showing an example of the arrangements of the pixel portion **104** and column A/D converter **105** according to the embodiment. At the pixel portion **104**, the plurality of pixels **201** are arranged. Also, at the pixel portion **104**, a plurality of output lines **202** are so arranged as to assign two or more output lines **202** for each of a plurality of columns on which the pixels **201** are aligned. In the arrangement shown in FIG. 2, four output lines are arranged for each column. The output lines **202** are connected to a constant current source (not shown).

[0020] In the arrangement shown in FIG. 2, the pixel **201** on the first column and the first row is connected to c1_vl1 out of the output lines **202**, and the pixel **201** on the first column and the second row is connected to c1_vl2 out of the output lines **202**. Similarly, the pixels **201** are connected to corresponding output lines **202** in a cycle of four rows. In FIG. 2, four output lines

202 connected to the pixels **201** on the first column are expressed as **c1_vl#** (#: 1 to 4), and four output lines **202** connected to the pixels **201** on the mth column are expressed as **cm_vl#**.

[0021] The output lines **202** are connected to the column A/D converter **105**. In the column A/D converter **105**, signal holding units **203** and converters **204** are arranged for each of a plurality of columns on which the pixels **201** are aligned. Each converter **204** converts an analog signal output from the corresponding signal holding unit **203** into a digital signal.

[0022] As shown in FIG. 3, the signal holding unit **203** includes holding circuits **213** and a selection circuit **223**. Each holding circuit **213** receives signals from, out of the plurality of pixels **201**, the pixels **201** connected to the corresponding output lines **202** via, out of the plurality of output lines **202**, two or more output lines (four in the arrangement shown in FIG. 2) assigned to a column on which the holding circuit **213** is arranged. The selection circuit **223** selects analog signals held by the holding circuits **213** and outputs them to the converter **204**.

[0023] In the embodiment, the converters **204** are arranged one by one on respective columns. Each converter **204** A/D-converts an analog signal output from the selection circuit **223** out of analog signals output to the signal holding unit **203** via the output lines **202** (in the case of the first pixel column, **c1_vl1**, **c1_vl2**, **c1_vl3**, and **c1_vl4**). For example, A/D conversion is repeated four times to convert analog signals output from the pixels **201** arranged on the first column and the first to fourth rows into digital signals. Signals output from the converter **204** are supplied to the data processor **106** via a signal line **adout_cm**. Note that a signal output from the converter **204** is digital data and is a signal of a plurality of bits.

[0024] FIG. 4 is a circuit diagram showing an example of the arrangement of the pixel **201**. Each of the pixels **201** arranged at the pixel portion **104** includes a photoelectric converter **311** and a photoelectric converter **313**. Further, the pixel **201** includes a charge transfer unit **312** that transfers charges accumulated in the photoelectric converter **311**, and a charge transfer unit **314** that transfers charges accumulated in the photoelectric converter **313**. The pixel **201** also includes a floating diffusion portion FD, a reset portion **315**, a signal amplifier **316**, and a selector **317**.

[0025] The photoelectric converters **311** and **313** convert incident light into charges. For example, the photoelectric converters **311** and **313** can be constituted using elements such as photodiodes.

[0026] The charge transfer unit **312** is arranged on an electrical path between the photoelectric converter **311** and the floating diffusion portion FD. The charge transfer unit **314** is arranged on an electrical path between the photoelectric converter **313** and the floating diffusion portion FD. The charge transfer unit **312** can be a transfer transistor that reads out charges accumulated in the photoelectric converter **311**. Conduction/non-conduction (ON/OFF) of the charge transfer unit **312** is controlled by a pixel transfer signal PTXA. Similarly, the charge transfer unit **314** can be a transfer transistor that reads out charges accumulated in the photoelectric converter **313**.

Conduction/non-conduction (ON/OFF) of the charge transfer unit **314** is controlled by a pixel transfer signal PTXB.

[0027] The reset portion **315** is arranged on an electrical path between the floating diffusion portion FD and a wiring pattern to which a power supply voltage VDD is supplied. The reset portion **315** can be a reset transistor that supplies the power supply voltage VDD to the floating diffusion portion FD to reset the potential of the floating diffusion portion FD. Conduction/non-conduction (ON/OFF) of the reset portion **315** is controlled by a pixel portion reset signal PRES.

[0028] The gate of the signal amplifier **316** is connected to the floating diffusion portion FD. Either of the drain and source of the signal amplifier **316** is connected to the power supply voltage VDD, and the other is connected to the selector **317**. The signal amplifier **316** is a source follower that converts charges of the floating diffusion portion FD into a voltage, and outputs the voltage to the output line **202**. The signal amplifier **316** can be constituted by, for example, a transistor.

[0029] The selector **317** is arranged on an electrical path between the output of the signal amplifier **316** and the output line **202**. The selector **317** is a transistor for selecting a pixel row that outputs a signal. Conduction/non-conduction (ON/OFF) of the selector **317** is controlled by a row selection

signal PSEL.

[0030] FIG. 5 is a schematic plan view of the pixel **201**. In each of the pixels **201** arranged at the pixel portion **104**, the photoelectric converters **311** and **313** share one microlens **318**. In other words, the photoelectric converters **311** and **313** are arranged in correspondence with one microlens **318**. That is, the photoelectric converters **311** and **313** receive light having passed through one microlens **318**.

[0031] FIG. 6 is a block diagram showing an example of the arrangement of the controller **102** according to the embodiment. The controller **102** can include a resistor controller **501**, control signal generators **502**, **505**, and **507**, and timing control counters **503**, **504**, and **506**. The resistor controller **501** holds various set values set by a control signal supplied from the CPU **101**. Based on a horizontal sync signal HD supplied from the CPU **101**, the control signal generator **502** generates a control signal that controls the timing control counters **503** and **504**. The timing control counters **503** and **504** operate based on the horizontal sync signal HD in accordance with the control signal supplied from the control signal generator **502**. The timing control counter **503** is used to control some of the holding circuits **213** in the signal holding unit **203**. The timing control counter **504** is used to control other holding circuits **213** in the signal holding unit **203**.

[0032] The control signal generator **505** generates a control signal that controls the signal holding unit **203**, based on the count values of the timing control counters **503** and **504** in accordance with a setting signal supplied from the resistor controller **501**. Also, the control signal generator **505** generates a signal that controls the timing control counter **506**. The timing control counter **506** operates based on the count values of the timing control counters **503** and **504** in accordance with a control signal supplied from the control signal generator **502**. The control signal generator **507** generates control signals that control the converter **204** and the data processor **106**, based on the count value of the timing control counter **506** in accordance with a setting signal supplied from the resistor controller **501**.

[0033] Next, a method of driving the photoelectric conversion device **100** according to the embodiment will be explained. FIGS. 7A to 7D are timing charts showing an example of the operation of the column A/D converter **105**. In the embodiment, signals of the pixels **201** arranged on four rows on each pixel column are simultaneously read out in the photoelectric conversion device **100**. The selection circuit **223** sequentially selects a signal output from one pixel **201** out of the simultaneously readout analog signals of the four pixels **201**, and outputs the signal to the converter **204**. The converter **204** A/D-converts the analog signals output from the selection circuit **223**, and sequentially outputs the signals for each pixel of each row. In the following description, as shown in FIG. 3, a circuit represented by SH1_# in the holding circuit **213** will be sometimes called a sample-and-hold circuit SH1, and a circuit represented by SH2_# will be sometimes called a sample-and-hold circuit SH2. The timing control counter **503** is used to control the sample-and-hold circuit SH1, and the timing control counter **504** is used to control the sample-and-hold circuit SH2, which will be described later.

[0034] At time T1, the CPU **101** supplies the horizontal sync signal HD to the controller **102**. The control signal generator **502** resets the timing control counter **503** to an initial value (for example, 0) based on the horizontal sync signal HD. At this time, the control signal generator **502** does not initialize the timing control counter **504**. At time T2, in accordance with the horizontal sync signal HD output from the CPU **101**, the control signal generator **502** resets the timing control counter **504** to an initial value (for example, 0) without initializing the timing control counter **503**. At time T3, in accordance with the horizontal sync signal HD output from the CPU **101**, the control signal generator **502** initializes the timing control counter **503** without initializing the timing control counter **504**. At time T4, in accordance with the horizontal sync signal HD output from the CPU **101**, the control signal generator **502** initializes the timing control counter **504** without initializing the timing control counter **503**. In this manner, the timing control counters **503** and **504** are alternately initialized and operate every time the horizontal sync signal HD is input.

[0035] At time $T1n$, the sample-and-hold circuit SH1 starts sampling the level (N signal) of the output line 202 before reading out signals from the photoelectric converters 311 and 313 of the pixel 201, and ends the sampling after a sample time Ts_{mpl_n} . Then, the row selector 103 causes four pixels 201 that are connected to four different output lines 202 arranged on each pixel column and arranged on different rows, to output analog signals based on the signal of the photoelectric converter 311. In response to this, at time $T1s1$, the sample-and-hold circuit SH1 starts sampling a focus detection signal S1 based on the signal of the photoelectric converter 311 that is read out from the pixel portion 104, and ends the sampling after a sample time Ts_{mpl_s1} . After the end of sampling the signal S1, the row selector 103 causes the four pixels 201 for which the signal S1 is sampled, to output analog signals based on the signals of the photoelectric converters 311 and 313. In response to this, at time $T1s2$, the sample-and-hold circuit SH1 starts sampling an image generation signal S2 based on the signals of the photoelectric converters 311 and 313 that are read out from the pixel portion 104, and ends the sampling after a sample time Ts_{mpl_s2} . The sample-and-hold circuit SH1 is in the hold state from time $T2$ to time $T3$, and holds the sampled signals.

[0036] At time $T3$, the horizontal sync signal HD is input, the timing control counter 503 is initialized, and the sample-and-hold circuit SH1 changes to the sample state again. At time $T3n$, the sample-and-hold circuit SH1 samples the signal S1 at time $T3s1$ and the signal S2 at time $T3s2$, and at time $T4$, changes to the hold state.

[0037] At time $T2n$, the sample-and-hold circuit SH2 starts sampling the level (N signal) of the output line 202 before reading out signals from the photoelectric converters 311 and 313 of the pixel 201, and ends the sampling after the sample time Ts_{mpl_n} . Then, the row selector 103 causes four pixels 201 that are connected to four different output lines 202 arranged on each pixel column and arranged on different rows, to output analog signals based on the signal of the photoelectric converter 311. For example, when the above-described sample-and-hold circuit SH1 samples signals output from the pixels 201 on the first to fourth rows, the sample-and-hold circuit SH2 can sample signals output from the pixels 201 on the fifth to eighth rows. In response to this, at time $T2s1$, the sample-and-hold circuit SH2 starts sampling the focus detection signal S1 based on the signal of the photoelectric converter 311 that is read out from the pixel portion 104, and ends the sampling after the sample time Ts_{mpl_s1} . After the end of sampling the signal S1, the row selector 103 causes the four pixels 201 (for example, the pixels 201 on the fifth to eighth rows) for which the signal S1 is sampled, to output analog signals based on the signals of the photoelectric converters 311 and 313. In response to this, at time $T2s2$, the sample-and-hold circuit SH2 starts sampling the image generation signal S2 based on the signals of the photoelectric converters 311 and 313 that are read out from the pixel portion 104, and ends the sampling after the sample time Ts_{mpl_s2} . The sample-and-hold circuit SH2 is in the hold state from time $T3$ to time $T4$, and holds the sampled signals.

[0038] When the sample-and-hold circuit SH1 is in the hold state, the sample-and-hold circuit SH2 performs sampling of signals, and when the sample-and-hold circuit SH2 is in the hold state, the sample-and-hold circuit SH1 performs sampling of signals. That is, the sample-and-hold circuits SH1 and SH2 alternately repeat the operation of sampling and holding signals every time the horizontal sync signal HD is input. Here, one cycle among cycles in which the horizontal sync signal HD is repetitively input will be called one horizontal period T_{hd} . In this case, the holding circuit 213 includes the sample-and-hold circuit SH1 that receives analog signals based on the signal of the photoelectric converter 311 and analog signals based on the signals of the photoelectric converters 311 and 313 in a given horizontal period, and the sample-and-hold circuit SH2 that receives analog signals based on the signal of the photoelectric converter 311 and analog signals based on the signals of the photoelectric converters 311 and 313 in a horizontal period subsequent to the given horizontal period.

[0039] At time $T2ad1$ when the value of the timing control counter 503 changes to a set value cnt_ad1 , the control signal generator 505 resets the timing control counter 506 to an initial value

(for example, 0). At the same time, the control signal generator **507** generates a signal by which the selection circuit **223** selects one sample-and-hold circuit from eight sample-and-hold circuits SH1_#. In accordance with the control signal generated by the control signal generator **507**, the selection circuit **223** selects a sample-and-hold circuit and outputs held analog signals to the converter **204**. The converter **204** converts the analog signals output from the signal holding unit **203** (selection circuit **223**) into digital signals.

[0040] Similarly, at time T2ad2 when the value of the timing control counter **503** changes to a set value cnt_ad2, the control signal generator **505** resets the timing control counter **506** to the initial value (for example, 0). Also, the control signal generator **507** generates a signal by which the selection circuit **223** selects one sample-and-hold circuit from the eight sample-and-hold circuits SH1_#. Then, the selection circuit **223** selects one sample-and-hold circuit from the sample-and-hold circuits SH1_#, and outputs held analog signals to the converter **204**. The converter **204** performs A/D conversion. This also applies to time T2ad3 to time T2ad8.

[0041] In the timing charts shown in FIGS. 7A to 7D, at time T2ad1, the selection circuit **223** selects the sample-and-hold circuit SH1_1 (S1). In response to this, the focus detection signal S1 output to cm_vl1 out of the output lines **202** is output to the converter **204** and A/D-converted. Similarly, the focus detection signal S1 output to cm_v12 out of the output lines **202** is selected at time T2ad2, the focus detection signal S1 output to cm_v13 out of the output lines **202** is selected at time T2ad3, and the focus detection signal S1 output to cm_v14 out of the output lines **202** is selected at time T2ad4. The focus detection signals S1 are respectively A/D-converted.

[0042] Then, at time T2ad5, the selection circuit **223** selects the sample-and-hold circuit SH1_1 (S2). In response to this, the image generation signal S2 output to cm_vl1 out of the output lines **202** is output to the converter **204** and A/D-converted. Similarly, the image generation signal S2 output to cm_v12 out of the output lines **202** is selected at time T2ad6, the image generation signal S2 output to cm_v13 out of the output lines **202** is selected at time T2ad7, and the image generation signal S2 output to cm_v14 out of the output lines **202** is selected at time T2ad8. The image generation signals S2 are respectively A/D-converted.

[0043] At this time, the sample-and-hold circuit SH2 performs sampling of signals and holds them. That is, every time the horizontal sync signal HD is input, the converter **204** switches signals held by the sample-and-hold circuits SH1 and SH2, and A/D-converts them. In a given horizontal period Thd, the selection circuit **223** selects analog signals supplied to the sample-and-hold circuit SH1 in a horizontal period Thd immediately preceding the given horizontal period Thd, and outputs them to the converter **204**. In a horizontal period Thd succeeding the horizontal period Thd, the selection circuit **223** selects analog signals supplied to the sample-and-hold circuit SH2 in the horizontal period Thd, and outputs them to the converter **204**. In the timing charts shown in FIGS. 7A to 7D, the converter **204** performs A/D conversion of the focus detection signal S1 four times and A/D conversion of the image generation signal S2 four times, that is, performs A/D conversion a total of eight times in time division.

[0044] The data processor **106** performs black level correction and the like on the A/D-converted digital signals, and outputs them to the signal output unit **107**. In the above-described way, data of four rows that are simultaneously read out in a given horizontal period Thd are A/D-converted for each row, and output for each pixel of each row from the photoelectric conversion device **100** via the signal output unit **107**.

[0045] Next, a method of driving the photoelectric conversion device **100** according to the embodiment will be explained with reference to FIGS. 8A to 8D. FIGS. 8A to 8D are timing charts showing an example of the operation of the column A/D converter **105** when some of the focus detection signals S1 are not A/D-converted. The operation of the holding circuit **213** (sample-and-hold circuits SH1 and SH2) in the signal holding unit **203** is the same as the operation shown in FIGS. 7A to 7D. However, the operation shown in FIGS. 8A to 8D is different from the operation shown in FIGS. 7A to 7D in the operation of the selection circuit **223** in the signal holding unit **203**.

The following description focuses on the difference, and a description of similar points will be omitted properly.

[0046] Similar to the operation shown in FIGS. 7A to 7D, at time T1 and time T2, the sample-and-hold circuit SH1 samples the focus detection signal S1 serving as an analog signal based on the signal of the photoelectric converter 311, and the image generation signal S2 serving as an analog signal based on the signals of the photoelectric converters 311 and 313. At time T2, the sample-and-hold circuit SH1 changes to the hold state.

[0047] At time T2ad1 when the timing control counter 503 changes to the set value cnt_ad1, the control signal generator 507 generates a signal by which the selection circuit 223 selects the sample-and-hold circuit SH1_1 (S1). Thus, the selection circuit 223 selects the sample-and-hold circuit SH1_1 (S1) holding the focus detection signal S1 output to cm_vl1 out of the output lines 202, and outputs the signal to the converter 204. The converter 204 A/D-converts the signal held by the sample-and-hold circuit SH1_1 (S1). Similarly, at time T2ad3, the selection circuit 223 selects the sample-and-hold circuit SH1_3 (S1) holding the focus detection signal S1 output to cm_vl3 out of the output lines 202, and outputs the signal to the converter 204. The converter 204 A/D-converts the signal held by the sample-and-hold circuit SH1_3 (S1).

[0048] Similar to the operation shown in FIGS. 7A to 7D, at time T2ad5, the selection circuit 223 selects the sample-and-hold circuit SH1_1 (S2). Then, the image generation signal S2 output to cm_vl1 out of the output lines 202 is output to the signal to the converter 204 and A/D-converted. Similarly, the image generation signal S2 output to cm_vl2 out of the output lines 202 is selected at time T2ad6, the image generation signal S2 output to cm_vl3 out of the output lines 202 is selected at time T2ad7, and the image generation signal S2 output to cm_vl4 out of the output lines 202 is selected at time T2ad8. The image generation signals S2 are respectively A/D-converted.

[0049] As for a signal sampled by the sample-and-hold circuit SH2, an operation similar to that for a signal sampled by the sample-and-hold circuit SH1 is performed, as shown in FIGS. 8A to 8D. In the timing charts shown in FIGS. 8A to 8D, A/D conversion of the focus detection signal S1 is performed two times during one horizontal period Thd, and A/D conversion of the image generation signal S2 is performed four times. That is, A/D conversion is performed a total of six times in one horizontal period Thd.

[0050] The operation of the holding circuit 213 (sample-and-hold circuits SH1 and SH2) in the signal holding unit 203, and the operation of the converter 204 every A/D conversion are the same. However, the number of focus detection analog signals (signals S1) output by the selection circuit 223 to the converter 204 is smaller than that of focus detection analog signals (signals S1) supplied to the holding circuit 213. Hence, the number of focus detection analog signals (signals S1) converted into digital signals by the converter 204 becomes smaller than that of focus detection analog signals (signals S1) supplied to the holding circuit 213. At this time, as shown in FIGS. 8A to 8D, the number of image generation analog signals output by the selection circuit 223 to the converter 204 may be equal to that of image generation analog signals supplied to the holding circuit 213. As a result, the number of times of A/D conversion in the converter 204 becomes smaller than that in the operation shown in FIGS. 7A to 7D, reducing the power consumption of the photoelectric conversion device 100 in one horizontal period Thd.

[0051] In the operation shown in FIGS. 8A to 8D, the converter 204 stands by at the timing of A/D conversion of the focus detection signal S1 thinned out from the operation shown in FIGS. 7A to 7D. However, the selection circuit 223 may select the sample-and-hold circuit SH1_3 (S1) at time T2ad2. Alternatively, for example, the selection circuit 223 may select the sample-and-hold circuit SH1_1 (S1) at time T2ad2, and select the sample-and-hold circuit SH1_3 (S1) at time T2ad3. Alternatively, for example, the selection circuit 223 may select the sample-and-hold circuit SH1_1 (S1) at time T2ad3, and select the sample-and-hold circuit SH1_3 (S1) at time T2ad4. Further, for example, the selection circuit 223 may select the sample-and-hold circuit SH1_1 (S1) at time T2ad1, and select the sample-and-hold circuit SH1_3 (S1) at time T2ad4.

[0052] The photoelectric conversion device **100** according to the embodiment includes the above-mentioned signal holding unit **203**. This can implement the same operation in vertical scanning of the pixels **201** regardless of whether which of the operation shown in FIGS. **8A** to **8D** in which A/D conversion of the focus detection signal **S1** is reduced, and the operation shown in FIGS. **7A** to **7D** in which all the focus detection signals **S1** are A/D-converted is performed. That is, the operation of the row selector **103** and that of the holding circuit **213** (sample-and-hold circuits **SH1** and **SH2**) do not change between the operation shown in FIGS. **7A** to **7D** and the operation shown in FIGS. **8A** to **8D**. Therefore, the time for each output line **202** until signals from the pixels **201** are output to the converter **204** after readout to the holding circuit **213** (sample-and-hold circuits **SH1** and **SH2**) does not change regardless of whether to thin out A/D conversion of the focus detection signal **S1**.

[0053] For example, a case where the signal holding unit **203** is not arranged and the time until signals from the pixels **201** are A/D-converted after readout changes depending on the operation will be considered. For example, when the signals **S1** output to the second and third output lines **202** out of signals output to four output lines **202** are thinned out, the signal **S1** of the fourth output line **202** may be sent to the converter **204** immediately after A/D conversion of the signal **S1** of the first output line **202**. To the contrary, when the signals **S1** are not thinned out, the signal **S1** of the fourth output line **202** is sent to the converter **204** after the end of A/D conversion of the signals **S1** of the first to third output lines **202**, and the time until the signal **S1** is A/D-converted after output to the output line **202** changes. In a case where the holding circuit **213** (sample-and-hold circuits **SH1** and **SH2**) is not arranged, the precision of A/D conversion may decrease owing to a temporal change of the signal level caused by leakage of the output line **202**. To the contrary, in the embodiment, the signal holding unit **203** can suppress such a decrease in A/D conversion precision.

[0054] As described above, the operation of the row selector **103** and that of the holding circuit **213** (sample-and-hold circuits **SH1** and **SH2**) do not change between the operation shown in FIGS. **7A** to **7D** and the operation shown in FIGS. **8A** to **8D**. When the operation shown in FIGS. **7A** to **7D** and the operation shown in FIGS. **8A** to **8D** are switched and used in accordance with the focus detection precision or the like, the controller **102** can control the operation of the row selector **103** and that of the holding circuit **213** using the same control signal. That is, a control signal output from the controller **102** can be easily designed.

[0055] In the operation shown in FIGS. **8A** to **8D**, A/D conversion of half of the focus detection signals **S1** is thinned out. However, the embodiment is not limited to this, and one signal **S1** or three signals **S1** may be thinned out. It only suffices to thin out a proper number of signals **S1** in accordance with shooting conditions, a precision necessary for focus detection, or the like. The photoelectric conversion device **100** may be so constituted as to appropriately set the number (ratio) of signals **S1** to be thinned out in accordance with shooting conditions, a precision necessary for focus detection, or the like. The number (ratio) of signals **S1** to be thinned out may be properly set by the user, or automatically set in accordance with shooting conditions or the like.

[0056] As described above, in the photoelectric conversion device **100** according to the embodiment, proper signals out of signals held by the holding circuit **213** (sample-and-hold circuits **SH1** and **SH2**) can be selected by the selection circuit **223** and output to the converter **204**. This can implement not only high-speed shooting, but also reduction of power consumption by decreasing the number of times of A/D conversion. Selecting proper signals can suppress a decrease in focus detection precision. As a result, the performance of the photoelectric conversion device **100** improves.

[0057] In the operation shown in FIGS. **8A** to **8D**, the holding circuit **213** (sample-and-hold circuits **SH1** and **SH2**) holds all the supplied focus detection signals **S1**, and the selection circuit **223** selects some of the focus detection signals **S1** held by the holding circuit **213** and outputs them to the converter **204**. However, the embodiment is not limited to this. For example, the holding circuit **213** may hold some of the supplied focus detection signals **S1**, and the selection circuit **223** may select the held signals **S1** and supply them to the converter **204**. For example, when analog signals

based on the signals of the photoelectric converters **311** of the pixels **201** are output to the output lines **202**, the holding circuit **213** (sample-and-hold circuits SH1 and SH2) need not hold signals to be thinned out.

[0058] Next, a modification of the operation shown in FIGS. **8A** to **8D** will be explained with reference to FIGS. **9A** to **9D**. FIGS. **9A** to **9D** are timing charts showing an example of the operation of the column A/D converter **105** when some of the focus detection signals **S1** are not A/D-converted, similar to the operation shown in FIGS. **8A** to **8D**. The operation of the holding circuit **213** (sample-and-hold circuits SH1 and SH2) in the signal holding unit **203** is the same as those shown in FIGS. **7** and **8**. However, the operation shown in FIGS. **9A** to **9D** is different from the operation shown in FIGS. **8A** to **8D** in the operation of the selection circuit **223** in the signal holding unit **203**. The following description focuses on the difference, and a description of similar points will be omitted properly.

[0059] In the operation shown in FIGS. **8A** to **8D**, the total numbers of focus detection signals **S1** and image generation signals **S2** output by the selection circuit **223** to the converter **204** in a given horizontal period T_{hd} are equal to those of focus detection signals **S1** and image generation signals **S2** output by the selection circuit **223** to the converter **204** in a horizontal period T_{hd} subsequent to the given horizontal period T_{hd} . In other words, the total numbers of focus detection signals **S1** and image generation signals **S2** that are selected by the selection circuit **223** from signals held by the sample-and-hold circuit SH1 and output to the converter **204** are equal to those of focus detection signals **S1** and image generation signals **S2** that are selected by the selection circuit **223** from signals held by the sample-and-hold circuit SH2 and output to the converter **204**. That is, the same numbers of focus detection signals **S1** and image generation signals **S2** are supplied to the converter **204** every horizontal period T_{hd} .

[0060] In contrast, in the operation shown in FIGS. **9A** to **9D**, the total numbers of focus detection signals **S1** and image generation signals **S2** output by the selection circuit **223** to the converter **204** in a given horizontal period T_{hd} are different from those of focus detection signals **S1** and image generation signals **S2** output by the selection circuit **223** to the converter **204** in a horizontal period T_{hd} subsequent to the given horizontal period T_{hd} . That is, different numbers of focus detection signals **S1** and image generation signals **S2** are supplied to the converter **204** every horizontal period T_{hd} (for example, alternately).

[0061] Even in the operation shown in FIGS. **9A** to **9D**, similar to the operation shown in FIGS. **7A** to **7D**, the sample-and-hold circuit SH1 samples at time $T1$ and time $T2$ the focus detection signal **S1** serving as an analog signal based on the signal of the photoelectric converter **311** and the image generation signal **S2** serving as an analog signal based on the signals of the photoelectric converters **311** and **313**. Then, the sample-and-hold circuit SH1 changes to the hold state at time $T2$.

[0062] At time $T2ad5$ when the timing control counter **503** changes to the set value cnt_ad5 , the control signal generator **507** generates a signal by which the selection circuit **223** selects the sample-and-hold circuit SH1_1 (**S2**). In response to this, the selection circuit **223** selects the sample-and-hold circuit SH1_1 (**S2**) holding the image generation signal **S2** output to cm_vl1 out of the output lines **202**, and outputs the signal to the converter **204**. The converter **204** A/D-converts the signal held by the sample-and-hold circuit SH1_1 (**S2**). Similarly, the image generation signal **S2** output to cm_vl2 out of the output lines **202** is selected at time $T2ad6$, the image generation signal **S2** output to cm_vl3 out of the output lines **202** is selected at time $T2ad7$, and the image generation signal **S2** output to cm_vl4 out of the output lines **202** is selected at time $T2ad8$. The image generation signals **S2** are respectively A/D-converted.

[0063] In this fashion, the selection circuit **223** does not output the focus detection signal **S1** to the converter **204** at time $T2$ and time $T3$. The focus detection signal **S1** supplied to the sample-and-hold circuit SH1 is not A/D-converted by the converter **204**.

[0064] At time $T2$ and time $T3$, the sample-and-hold circuit SH2 samples the focus detection signal **S1** serving as an analog signal based on the signal of the photoelectric converter **311** and the image

generation signal **S2** serving as an analog signal based on the signals of the photoelectric converters **311** and **313**. At time **T3**, the sample-and-hold circuit **SH2** changes to the hold state.

[0065] At time **T3ad1** when the timing control counter **504** changes to the set value **cnt_ad1**, the selection circuit **223** selects the sample-and-hold circuit **SH2_1** (**S1**). In response to this, the focus detection signal **S1** output to **cm_vl1** out of the output lines **202** is output to the converter **204** and A/D-converted. Similarly, the focus detection signal **S1** output to **cm_v12** out of the output lines **202** is selected at time **T3ad2**, the focus detection signal **S1** output to **cm_v13** out of the output lines **202** is selected at time **T3ad3**, and the focus detection signal **S1** output to **cm_v14** out of the output lines **202** is selected at time **T3ad4**. The focus detection signals **S1** are respectively A/D-converted.

[0066] Then, at time **T3ad5** when the timing control counter **504** changes to the set value **cnt_ad5**, the selection circuit **223** selects the sample-and-hold circuit **SH2_1** (**S2**). Then, the image generation signal **S2** output to **cm_vl1** out of the output lines **202** is output to the converter **204** and A/D-converted. Similarly, the image generation signal **S2** output to **cm_v12** out of the output lines **202** is selected at time **T3ad6**, the image generation signal **S2** output to **cm_v13** out of the output lines **202** is selected at time **T3ad7**, and the image generation signal **S2** output to **cm_v14** out of the output lines **202** is selected at time **T3ad8**. The image generation signals **S2** are respectively A/D-converted.

[0067] In this manner, at time **T3** and time **T4**, the selection circuit **223** outputs to the converter **204** all the focus detection signals **S1** and image generation signals **S2** held by the sample-and-hold circuit **SH2**. Hence, the converter **204** A/D-converts all the focus detection signals **S1** and the image generation signals **S2** that are supplied to the sample-and-hold circuit **SH2**. In the hold period between time **T3** and time **T4**, the converter **204** A/D-converts all the focus detection signals (**S1**) and the image generation signals (**S2**).

[0068] In the operation shown in FIGS. **9A** to **9D**, only the image generation signals **S2** are A/D-converted without A/D-converting the focus detection signals **S1** supplied to the sample-and-hold circuit **SH1**. The focus detection signals **S1** and the image generation signals **S2** that are supplied to the sample-and-hold circuit **SH2** are A/D-converted. In the operation shown in FIGS. **8A** to **8D**, whether to A/D-convert the focus detection signal **S1** is switched between the hold period of the sample-and-hold circuit **SH1** and that of the sample-and-hold circuit **SH2**.

[0069] In the operation shown in FIGS. **9A** to **9D**, A/D conversion of the focus detection signal **S1** is performed four times, and A/D conversion of the image generation signal **S2** is performed eight times in a period double the horizontal period **Thd**. That is, A/D conversion is performed a total of 12 times in a period double the horizontal period **Thd**. Since A/D conversion is performed a total of 16 times in a period double the horizontal period **Thd** in the operation shown in FIGS. **7A** to **7D**, the number of times of A/D conversion in the operation shown in FIGS. **9A** to **9D** is smaller than that in the operation shown in FIGS. **7A** to **7D**.

[0070] Even in the operation shown in FIGS. **9A** to **9D**, the operation of the holding circuit **213** (sample-and-hold circuits **SH1** and **SH2**) in the signal holding unit **203**, and the operation of the converter **204** every A/D conversion are the same. However, the number of focus detection analog signals (signals **S1**) output by the selection circuit **223** to the converter **204** is smaller than that of focus detection analog signals (signals **S1**) supplied to the holding circuit **213**. Thus, the number of focus detection analog signals (signals **S1**) converted into digital signals by the converter **204** becomes smaller than that of focus detection analog signals (signals **S1**) supplied to the holding circuit **213**. As a result, the number of times of A/D conversion in the converter **204** becomes smaller than that in the operation shown in FIGS. **7A** to **7D**, reducing the power consumption of the photoelectric conversion device **100**.

[0071] In the operation shown in FIGS. **9A** to **9D**, the focus detection signal **S1** supplied to the sample-and-hold circuit **SH1** is not A/D-converted. However, the embodiment is not limited to this, and the focus detection signal **S1** supplied to the sample-and-hold circuit **SH1** may be A/D-

converted without A/D-converting the focus detection signal **S1** supplied to the sample-and-hold circuit **SH2**.

[0072] Alternatively, some of the focus detection signals **S1** supplied to the sample-and-hold circuit **SH1** may be A/D-converted in accordance with shooting conditions, a precision necessary for focus detection, or the like. Similar to the operation shown in FIGS. **8A** to **8D**, it is only necessary to thin out a proper number of signals **S1** in accordance with shooting conditions, a precision necessary for focus detection, or the like.

[0073] Even in the operation shown in FIGS. **9A** to **9D**, the photoelectric conversion device **100** according to the embodiment can use the selection circuit **223** to select proper signals out of signals held by the holding circuit **213** (sample-and-hold circuits **SH1** and **SH2**) and output them to the converter **204**. This can implement not only high-speed shooting, but also reduction of power consumption by decreasing the number of times of A/D conversion. Selecting proper signals can suppress a decrease in focus detection precision. Therefore, the performance of the photoelectric conversion device **100** improves.

[0074] Although the plurality of output lines **202** are provided for one column of pixels in the embodiment, the arrangement is not limited to this. For example, only one output line **202** may be arranged for one column of pixels. Even in this case, the selection circuit **223** is provided to select some output lines **202** from the plurality of output lines **202**, that is, the plurality of output lines **202** arranged in correspondence with a plurality of columns of pixels, and output signals to the holding circuit **213**. Even in this form, the operations according to the embodiment respectively shown in FIGS. **7** to **9** can be performed.

[0075] An application example of the photoelectric conversion device **100** according to the embodiment will be explained with reference to FIG. **10**. FIG. **10** is a schematic view of an apparatus **9191** including the photoelectric conversion device **100**. As shown in FIG. **10**, the photoelectric conversion device **100** is housed in a package **920**. The package **920** can include a base to which the photoelectric conversion device **100** is fixed, and a lid such as glass facing the photoelectric conversion device **100**. The package **920** can further include joint members such as bonding wires and bumps that connect terminals provided on the base and pads provided on the photoelectric conversion device **100**.

[0076] The apparatus **9191** can include at least one of an optical device **940**, a control device **950**, a processing device **960**, a display device **970**, a storage device **980**, and a mechanical device **990**. The optical device **940** is implemented by, for example, a lens, a shutter, and a mirror. The control device **950** controls the photoelectric conversion device **100**. The control device **950** is, for example, a semiconductor device such as an ASIC.

[0077] The processing device **960** processes a signal output from the photoelectric conversion device **100**. The processing device **960** is a semiconductor device such as a CPU or an ASIC for forming an analog front end (AFE) or a digital front end (DFE). The display device **970** is an EL display device or a liquid crystal display device that displays information (image) obtained by the photoelectric conversion device **100**. The storage device **980** is a magnetic device or a semiconductor device that stores the information (image) obtained by the photoelectric conversion device **100**. The storage device **980** is a volatile memory such as an SRAM or a DRAM, or a nonvolatile memory such as a flash memory or a hard disk drive.

[0078] The mechanical device **990** includes a moving or propulsion unit such as a motor or an engine. In the apparatus **9191**, the signal output from the photoelectric conversion device **100** is displayed on the display device **970** or transmitted to an external device by a communication device (not shown) included in the apparatus **9191**. Hence, the apparatus **9191** may further include the storage device **980** and the processing device **960** in addition to the memory circuits and arithmetic circuits included in the photoelectric conversion device **100**. The mechanical device **990** may be controlled based on the signal output from the photoelectric conversion device **100**.

[0079] In addition, the apparatus **9191** is suitable for an electronic apparatus such as an information

terminal (for example, a smartphone or a wearable terminal) which has a shooting function or a camera (for example, an interchangeable lens camera, a compact camera, a video camera, or a monitoring camera). The mechanical device **990** in the camera can drive the components of the optical device **940** in order to perform zooming, an in-focus operation, and a shutter operation. Alternatively, the mechanical device **990** in the camera can move the photoelectric conversion device **100** in order to perform an anti-vibration operation.

[0080] Furthermore, the apparatus **9191** can also be applied to an onboard camera mounted in a transportation apparatus such as a vehicle, a ship, an airplane, or an industrial robot. The mechanical device **990** in the transportation apparatus can be used as a moving device. The apparatus **9191** as the transportation apparatus is suitable for a device that transports the photoelectric conversion device **100** or a device that uses an image capturing function to assist and/or automate driving (steering). The processing device **960** for assisting and/or automating driving (steering) can perform, based on the information obtained by the photoelectric conversion device **100**, processing for operating the mechanical device **990** as a moving device. The apparatus **9191** incorporating the photoelectric conversion device **100** can be widely applied to an apparatus using object recognition such as an intelligent transport system (ITS), in addition to the transportation apparatus. Alternatively, the apparatus **9191** may be a medical apparatus such as an endoscope, a measurement apparatus such as a distance measurement sensor, an analysis device such as an electron microscope, or an office apparatus such as a copy machine.

[0081] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0082] This application claims the benefit of Japanese Patent Application No. 2024-032522, filed Mar. 4, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A photoelectric conversion device comprising: a plurality of pixels arranged to constitute a plurality of rows and a plurality of columns; a plurality of output lines; and a row selector configured to select, from the plurality of pixels, a pixel that outputs a signal, wherein a signal holding unit, and a converter configured to convert an analog signal output from the signal holding unit into a digital signal are arranged for each column of the plurality of columns, the signal holding unit includes a holding circuit configured to receive signals from pixels connected to at least two output lines out of the plurality of pixels via, out of the plurality of output lines, the at least two output lines assigned to the column on which the signal holding unit is arranged, and a selection circuit, each of the plurality of pixels includes a first photoelectric converter and a second photoelectric converter, the row selector causes at least two pixels that are connected to different output lines out of the at least two output lines and arranged on different rows out of the plurality of rows, to output first analog signals based on a signal of the first photoelectric converter, and then output second analog signals based on a signal of the first photoelectric converter and a signal of the second photoelectric converter, the selection circuit selects the first analog signals and the second analog signals that are held by the holding circuit, and outputs the first analog signals and the second analog signals to the converter, and a number of first analog signals output by the selection circuit to the converter is smaller than a number of first analog signals supplied to the holding circuit.
2. The device according to claim 1, wherein the plurality of output lines are provided to assign at least two output lines to each column of the plurality of columns.
3. The device according to claim 1, wherein a number of second analog signals output by the selection circuit to the converter is equal to a number of second analog signals supplied to the

holding circuit.

4. The device according to claim 1, wherein a number of first analog signals converted into digital signals by the converter is smaller than a number of first analog signals supplied to the holding circuit.

5. The device according to claim 1, wherein the holding circuit includes a first holding circuit configured to receive the first analog signal and the second analog signal in a first horizontal period, and a second holding circuit configured to receive the first analog signal and the second analog signal in a second horizontal period subsequent to the first horizontal period, and in the second horizontal period, the selection circuit selects the first analog signal and the second analog signal that are supplied to the first holding circuit in the first horizontal period, and outputs the first analog signal and the second analog signal to the converter, and in a third horizontal period subsequent to the second horizontal period, selects the first analog signal and the second analog signal that are supplied to the second holding circuit in the second horizontal period, and outputs the first analog signal and the second analog signal to the converter.

6. The device according to claim 5, wherein total numbers of first analog signals and second analog signals that are output by the selection circuit to the converter in the second horizontal period are equal to total numbers of first analog signals and second analog signals that are output by the selection circuit to the converter in the third horizontal period.

7. The device according to claim 5, wherein total numbers of first analog signals and second analog signals that are output by the selection circuit to the converter in the second horizontal period are different from total numbers of first analog signals and second analog signals that are output by the selection circuit to the converter in the third horizontal period.

8. The device according to claim 7, wherein the selection circuit does not output the first analog signal to the converter in the second horizontal period.

9. The device according to claim 8, wherein in the third horizontal period, the selection circuit supplies, to the converter, all the first analog signals that are supplied to the second holding circuit in the second horizontal period.

10. The device according to claim 1, wherein the converter converts, into digital signals in time division, the first analog signal and the second analog signal that are output from the selection circuit.

11. The device according to claim 1, wherein the first analog signal is a focus detection signal, and the second analog signal is an image generation signal.

12. The device according to claim 1, wherein each of the plurality of pixels includes a microlens, and the first photoelectric converter and second photoelectric converter of each of the plurality of pixels are arranged in correspondence with one microlens.

13. The device according to claim 2, wherein each of the plurality of pixels includes a microlens, and the first photoelectric converter and second photoelectric converter of each of the plurality of pixels are arranged in correspondence with one microlens.

14. The device according to claim 1, wherein the holding circuit holds some of the supplied first analog signals, and the selection circuit selects the some of the first analog signals and supplies the some of the first analog signals to the converter.

15. The device according to claim 1, wherein the holding circuit holds all the supplied first analog signals, and the selection circuit selects some of the first analog signals held by the holding circuit, and supplies the some of the first analog signals to the converter.

16. An apparatus comprising: the photoelectric conversion device according to claim 1; and a processing device configured to process a signal output from the photoelectric conversion device.
